

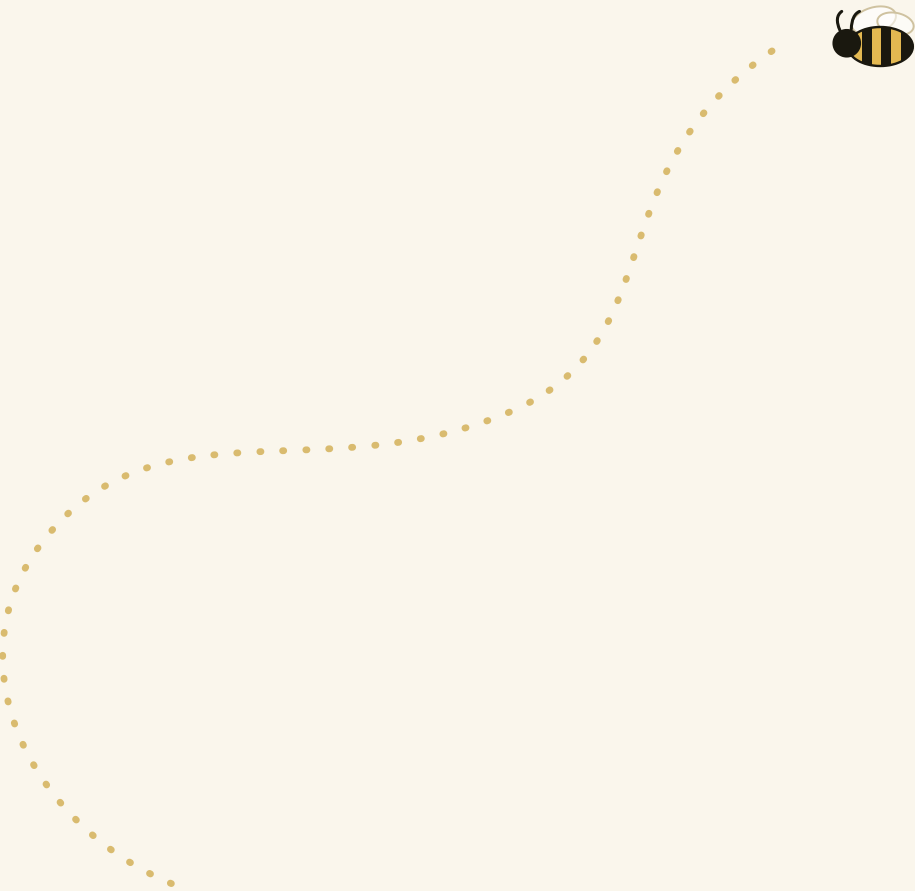


DISTRIBUTED ENERGY PLATFORM

The universal translator for the distributed grid.

One gateway. Any asset. Any application.

NORTH STAR — MEGAWATTS UNDER MANAGEMENT



THE PROBLEM

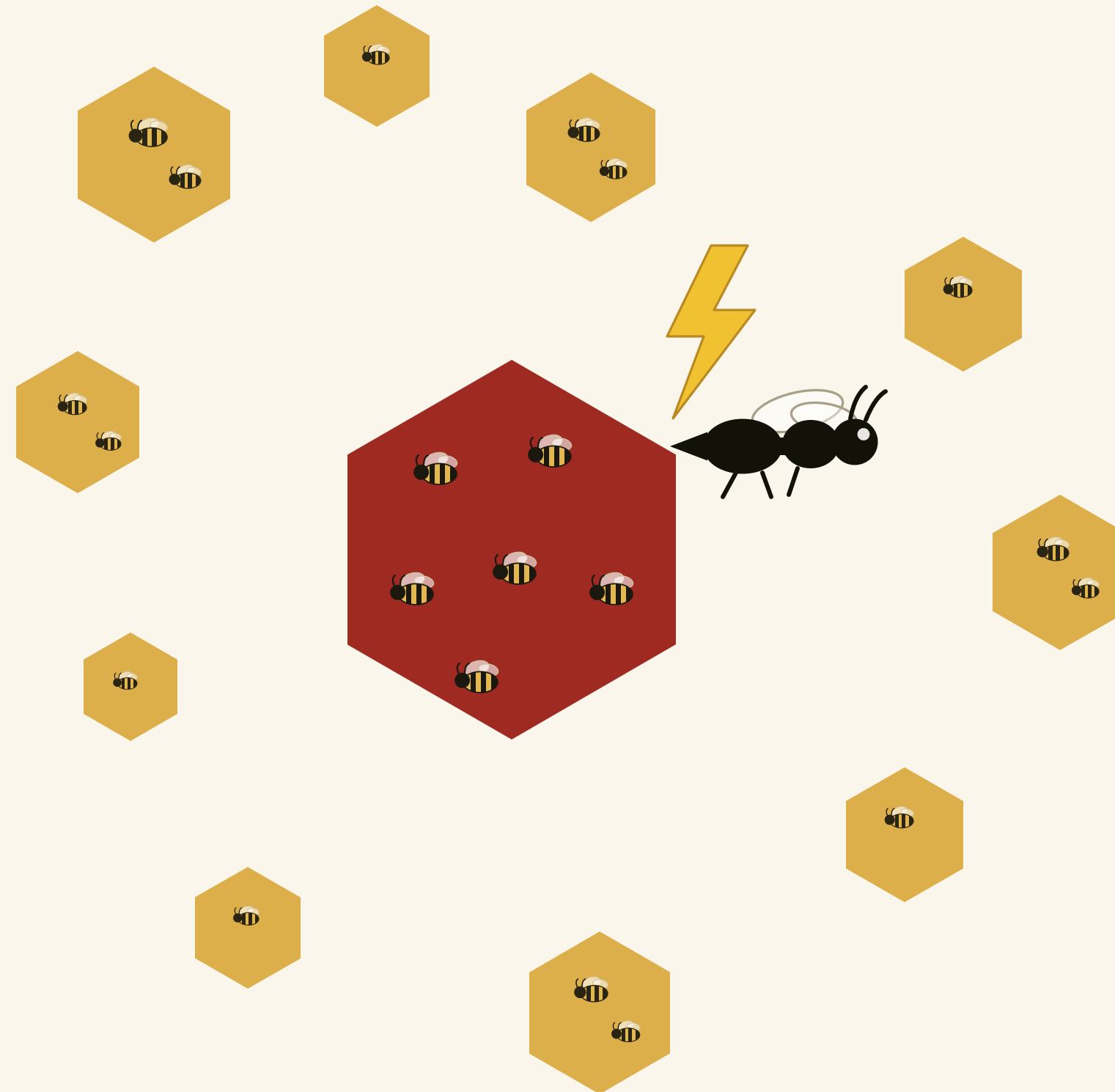
Today, the hive is fragmented.

Every OEM locks its devices in a separate silo. Over 100 million DERs are deployed worldwide with no common dispatch layer beneath them.

80%

of DER fleets still lack dispatchability. Existing APIs are too expensive, and lack interoperability and security.

Each silo was built correctly. None of them was built to be read by anyone else.



Two curves just crossed.

DER growth has outrun the infrastructure built to carry it, and interconnection rules just made dispatchability mandatory.



IEEE 1547 and IEEE 2030.5 are day one

They are the interconnection requirement for distribution-connected assets, and there is no cheap way to speak them at the edge.

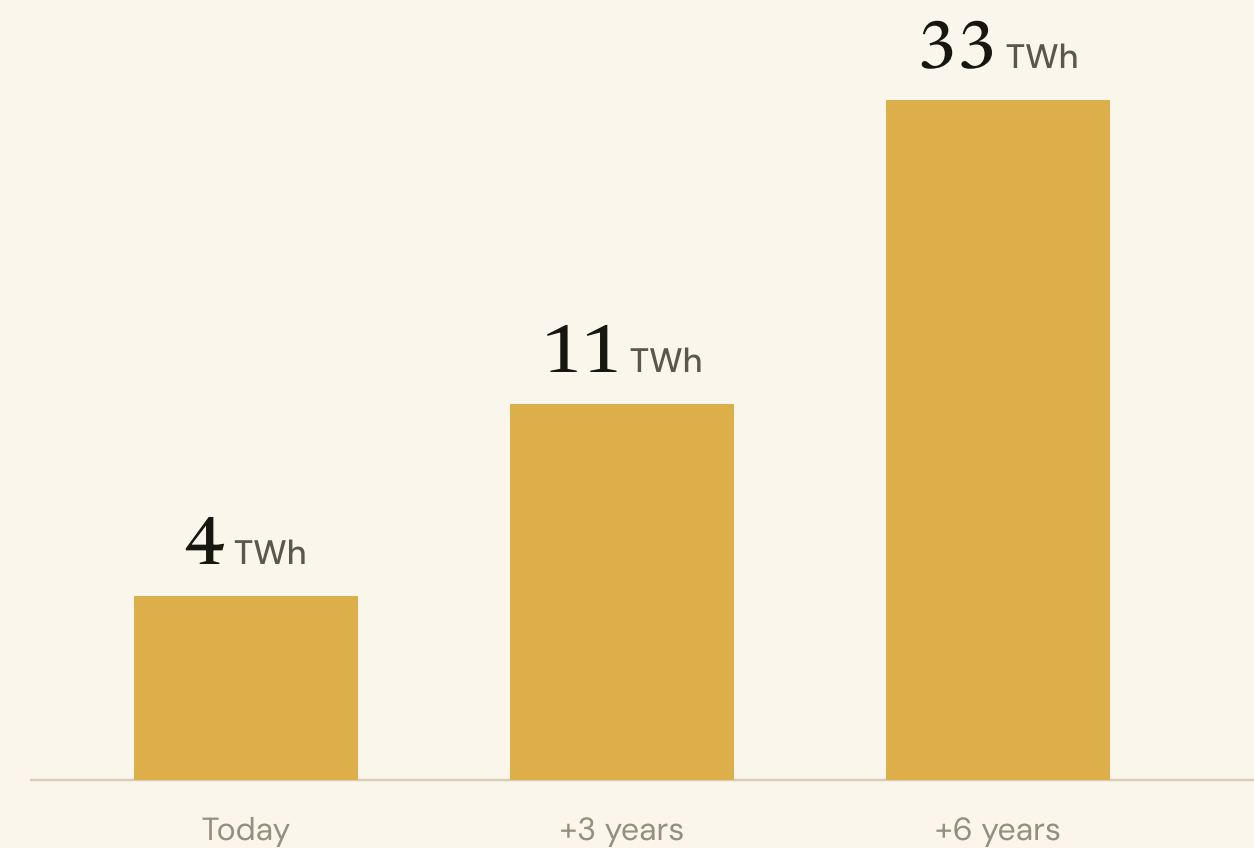


Edge compute got cheap enough to be everywhere

A gateway can now sit on every asset in the fleet, not only on the largest ones that justified an engineer's visit.

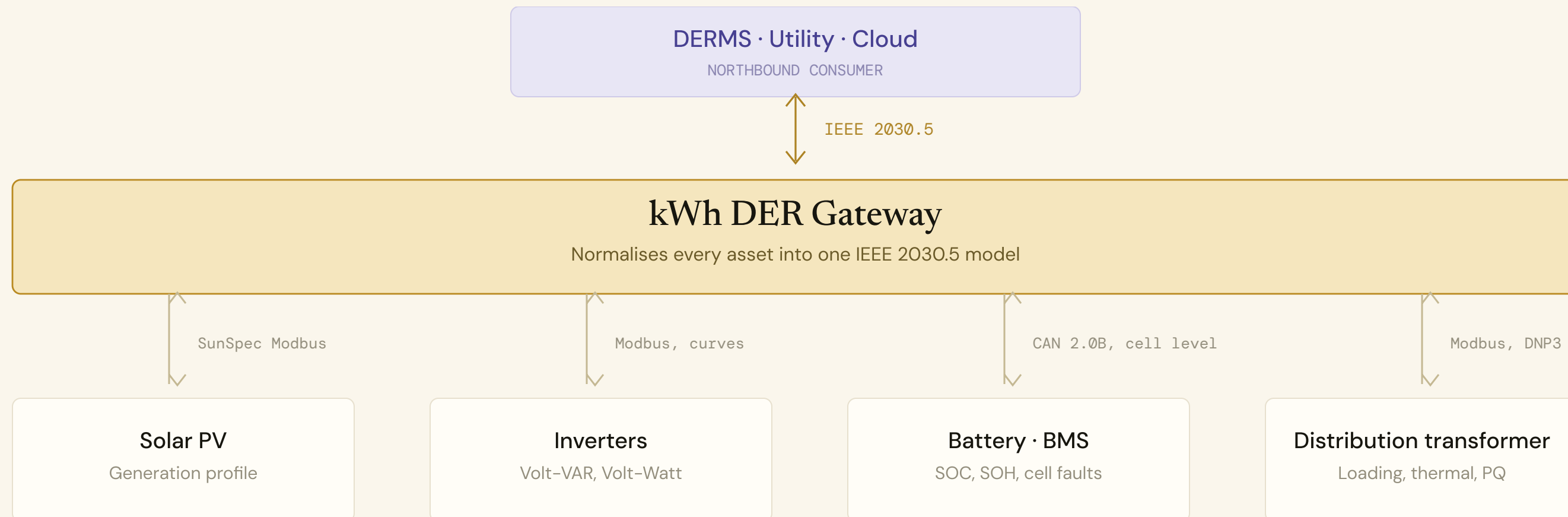
INSTALLED DER CAPACITY

Illustrative shape, not a forecast



One gateway. One common language.

The gateway performs **model** translation, not just protocol translation. Many dialects go in the bottom. One standard comes out the top.



The transformer is read through its existing metering or monitoring IED. Neither gateway model meters. Both read what is already measured.

THE SOLUTION • SECURITY

Every asset guards itself.

Each DER sits in its own sealed cell. One compromised device never reaches the fleet.



Trusted Execution Environment

Policy is enforced on the edge device, not in a cloud an attacker can reach.



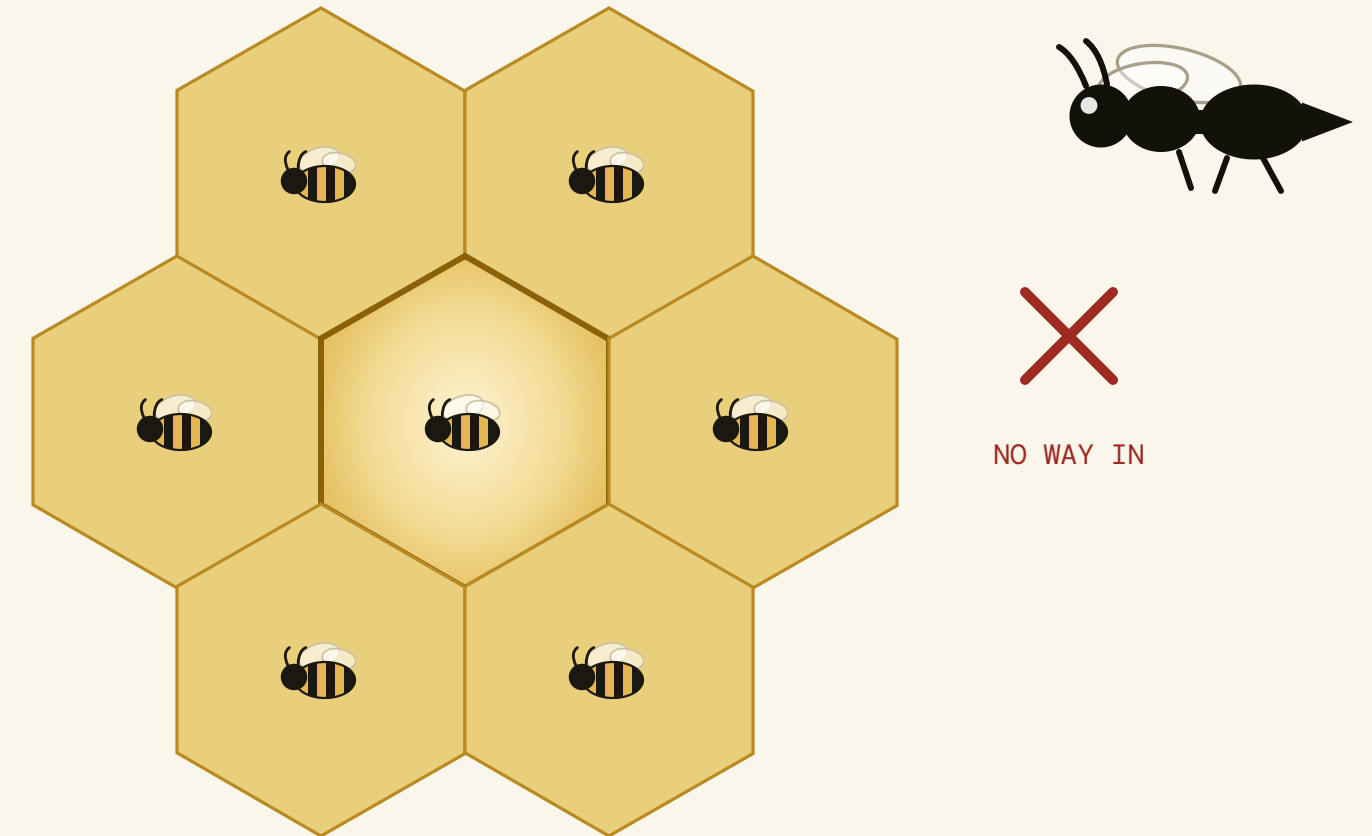
Zero-trust device identity

Each DER is enrolled and authenticated alone. No shared secret, no lateral move.



Data sovereignty

Telemetry stays in-jurisdiction. GDPR, India's DPDP Act and US state law by construction, not by policy document.



ONE CELL LIT • THE FLEET UNTOUCHED

An agent can plan a dispatch. Nothing lets it act.

There is no interface between a model that reasons and hardware that must not be wrong.

NO BOUNDED INTERFACE

An API key to a battery is not a safety model. Access is not the same as authority.

NO POLICY AT THE EDGE

If the link drops or the model errs, nothing on site stops the command from executing.

NO VERIFIABLE RECORD

An action nobody can audit is an action nobody will authorise. Utilities do not sign for a black box.

AGENT PROPOSES

MCP-native. The agent queries the hive and requests an action in the same language every asset already speaks.

BOUNDARY

THE CELL DISPOSES

Policy-as-code runs on the edge device. Every request is checked against that asset's safe operating envelope before anything moves.



The same lit cell that turns the wasp away bounds the agent. One security model, two threats, no extra engineering.

The agent can ask the hive for anything. **The cell decides what is safe.**

Two gateways.
One core.

Same compute, same stack, same northbound interface. The difference is where it has to survive.

	MODEL A – DER GATEWAY	MODEL B – DT GATEWAY
APPLICATION	BESS, solar PV, inverters	Distribution transformer sites
CLASS	SBC–tier embedded gateway	Ruggedised, cellular–connected
ENVIRONMENT	Indoor, enclosure–protected · IP20	Outdoor pole or pad mount · IP65 / IP66
TEMPERATURE	0 to +50 °C commercial	–40 to +85 °C
READS	Inverter and BMS telemetry	The transformer's existing meter or IED
SOUTHBOUND	RS–485, Ethernet, CAN 2.0B	RS–485, Ethernet, DNP3, optional IEC 61850
CONNECTIVITY	Ethernet, optional 4G LTE	Dual Ethernet + 4G LTE, GPS
TARGET PRICE AT VOLUME	~₹6,000	~₹24,000

Neither model meters. Both read what is already measured, and normalise it into one IEEE 2030.5 model.

Target BOM–driven prices at volume. Early certified, industrial–temperature units land higher.

What each side gets.



Asset owner

A stranded asset becomes a revenue-earning one.

Instead of waiting for a replacement they will not buy.



Utility · DISCOM

Visibility and control at the LV network edge.

Instead of a blind spot below the feeder.



OEM

Grid-program eligibility without building a standards stack.

Instead of twelve months of protocol engineering.



Financier

Independent telemetry on the whole book.

Instead of residual value they cannot verify.

One integration replaces a project per site.

The blind spot below the feeder, closed.

Transformer blindness is a problem a DISCOM already knows it has, and already budgets against.

MODEL A – AT THE DER

Visibility

Active power per site and per feeder · reactive power and power factor · availability and connection status · reverse power flow

Grid support

Volt-VAR and Volt-Watt curve status · voltage at point of connection · frequency response · ride-through event log

Dispatch

Curtailement requested vs delivered · setpoint compliance · ramp-rate adherence · response latency

Battery

State of charge and state of health · cycles · round-trip efficiency · cell-level temperature and fault flags

MODEL B – AT THE TRANSFORMER

Loading and health

kVA vs rating · overload events and duration · oil and winding temperature · estimated loss-of-life · tap position

Power quality, LV side

Per-phase voltage · phase imbalance · reverse power flow duration · sags, swells, interruptions

Loss and efficiency

Energy through the transformer per interval · input vs downstream consumption for AT&C loss localisation · overloaded vs underutilised DTs

Reliability

Outage and restoration timestamps · interruption duration and frequency feeding SAIDI and SAIFI · last-gasp reporting through power-loss ride-through

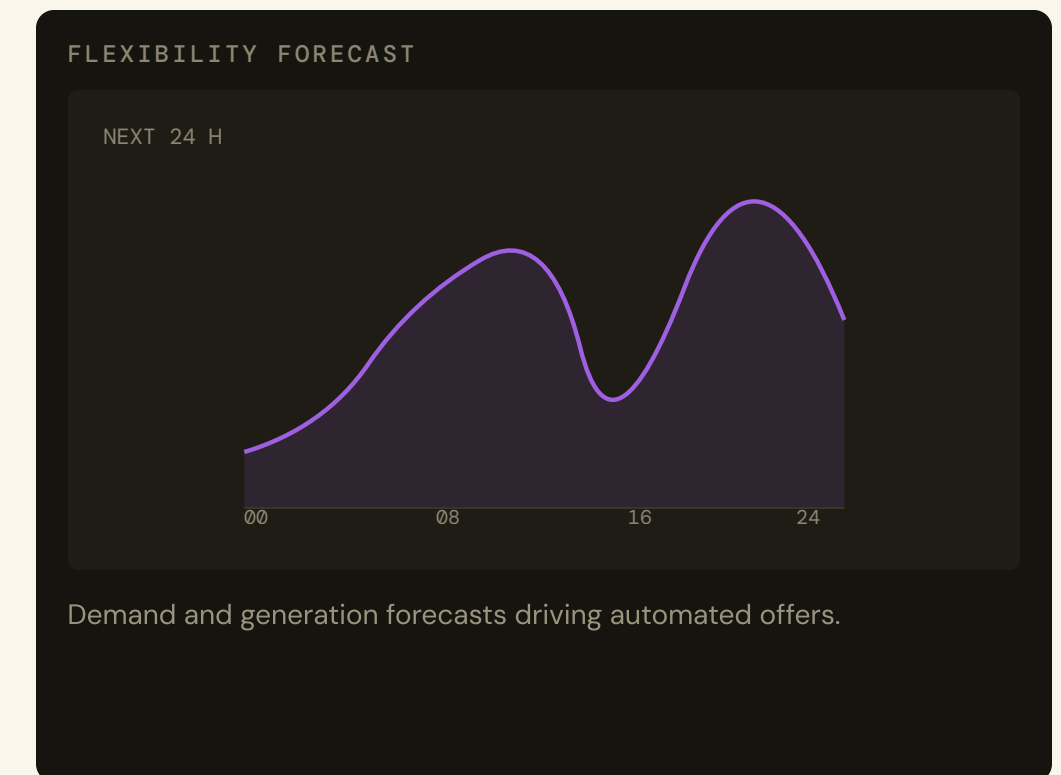
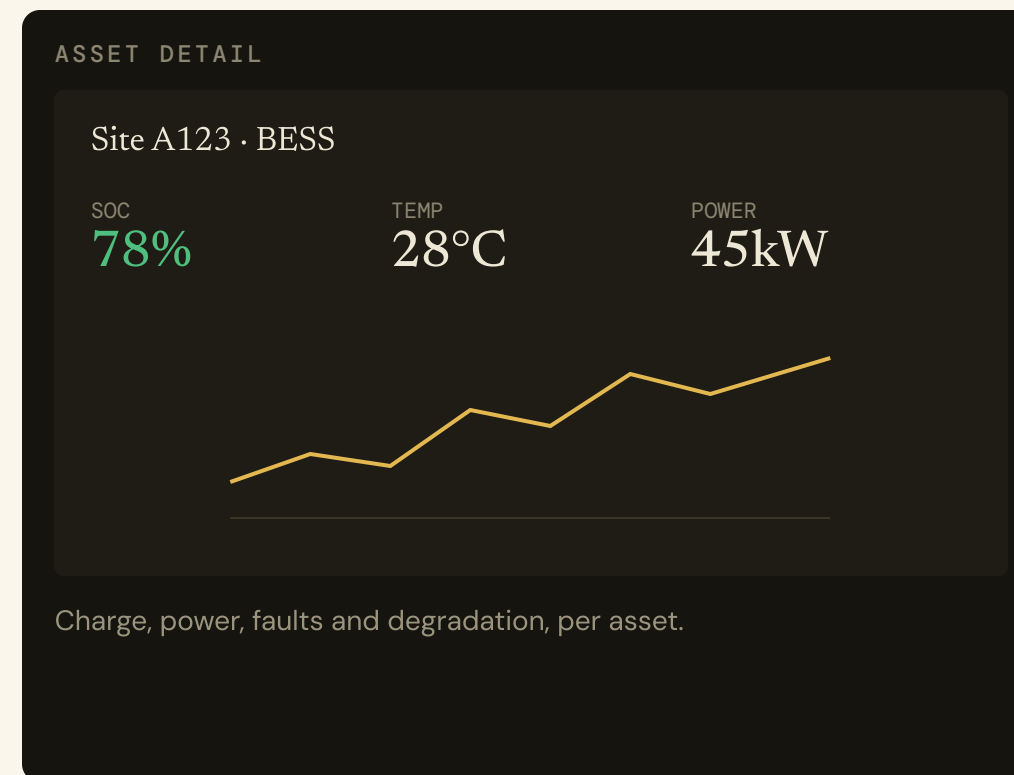
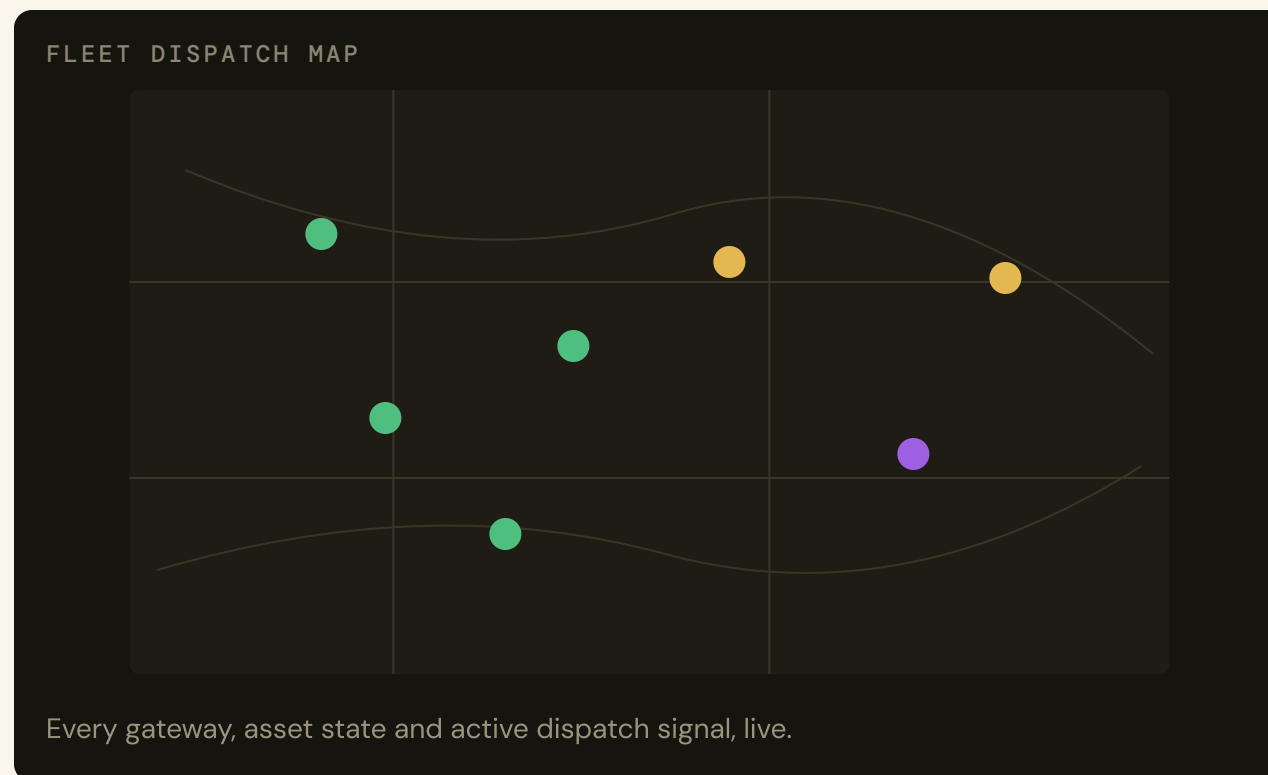
- Fewer transformer failures
- AT&C loss localisation
- SAIDI / SAIFI at fine grain
- Rooftop solar without LV overload
- Deferred transformer replacement

Transformer metrics are derived from the DT's existing instrumentation. Availability depends on what that system exposes.

What the operator actually sees.

340ms

Dispatch latency, measured live



kWh Fleet Console ● LIVE

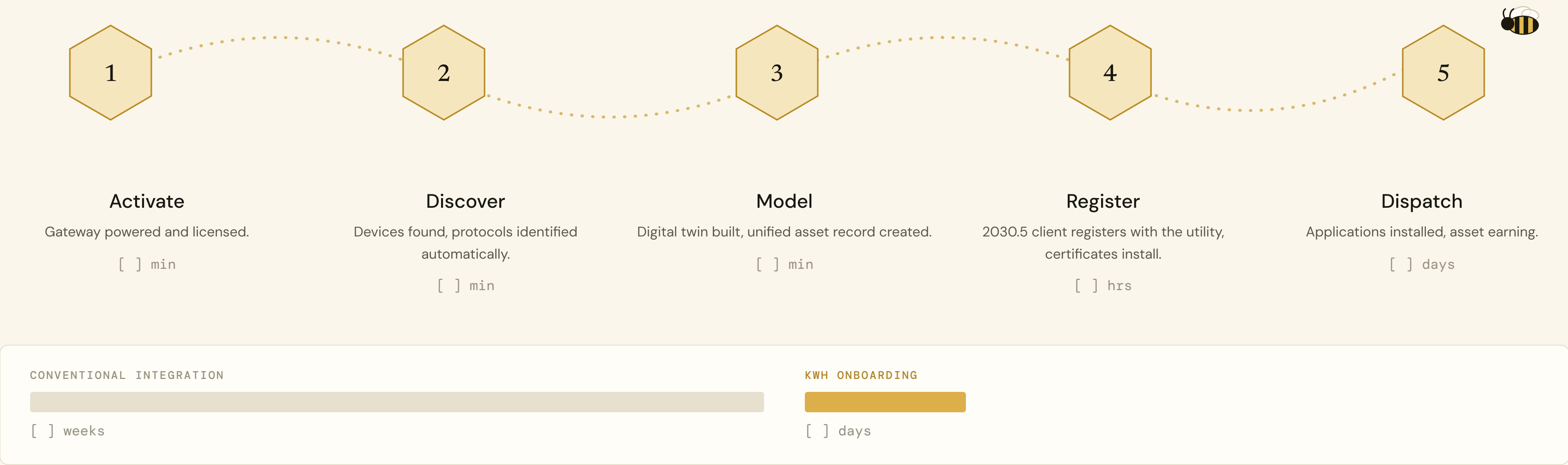
BESS 004 · Kintech Synergy

State of charge 78%

Dispatch Active

Latency 340ms

Live in [X] days, not a quarter.



No site survey per asset. No vendor engineering engagement. No custom driver written on site.

How the gateway reaches the asset.

Every route lands the same core. Only the enclosure and the party who pays for it change.



Custom PCB in fabrication with PCB Power India.

Stranded megawatts are the beachhead.

[X GW]

INSTALLED BASE

Behind the meter

[Y GW]

ADDRESSABLE STRANDED

Reachable via retrofit and OEM cloud

[\$Z]

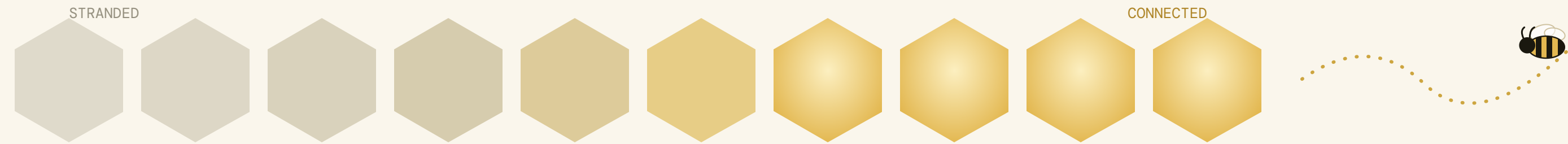
ANNUAL SAM

SaaS, hardware and revenue share

[X million]

DISTRIBUTION TRANSFORMERS

Today largely uninstrumented



Beachhead: India DISCOMs via the Beckn DEG channel, expanding to U.S. utilities and aggregators on IEEE 2030.5. India is the proving ground for velocity, not the market cap.

Model B is Model A plus the environment.

The four-times price gap is entirely explainable by physics, not by markup.

MODEL A – DER EDGE

~₹6,000

Indoor · IP20 · 0 to +50 °C

CONTRIBUTION PER UNIT

[₹] [%]

MODEL B – TRANSFORMER EDGE

~₹24,000

Outdoor · IP65/66 · -40 to +85 °C

CONTRIBUTION PER UNIT

[₹] [%]

WHERE THE DIFFERENCE GOES		
+	IP65 / IP66 outdoor enclosure	[₹]
+	-40 to +85 °C component grade	[₹]
+	Surge and EMC protection, pole-top	[₹]
+	Cellular modem and antenna	[₹]
+	Wider isolated I/O count	[₹]
+	Power-loss ride-through for last gasp	[₹]

BLENDED CONTRIBUTION	UNITS TO BREAKEVEN ON TOOLING
[₹]	[X]

Hardware is not the margin engine. It is the wedge that installs the software, and Model B is the wedge that gets a DISCOM to pay for the network.

Model B price composition to be confirmed as hardware-only versus hardware plus connectivity plus SLA.

The gateway installs the software.

WHAT SITS ON THE BOX

Applications · DERMS · third-party

kWh Platform — SaaS

Gateway OS

Edge agent — policy-as-code

Hardware

Hardware is sold once. Everything above it recurs.

Each device profile built unlocks a whole model class. The library compounds; the cost to onboard does not.

RECURRING ECONOMICS, PER ASSET

SaaS revenue per asset / month	[]
less Cloud and telemetry storage	[]
less Support and success	[]
Gross profit / asset / month	[] [%]

CAC / ASSET
[]

PAYBACK
[X] mo

LTV / CAC
[X]×

NET REV. RETENTION
[X%]

COST PER MW, VIA GATEWAY

[]

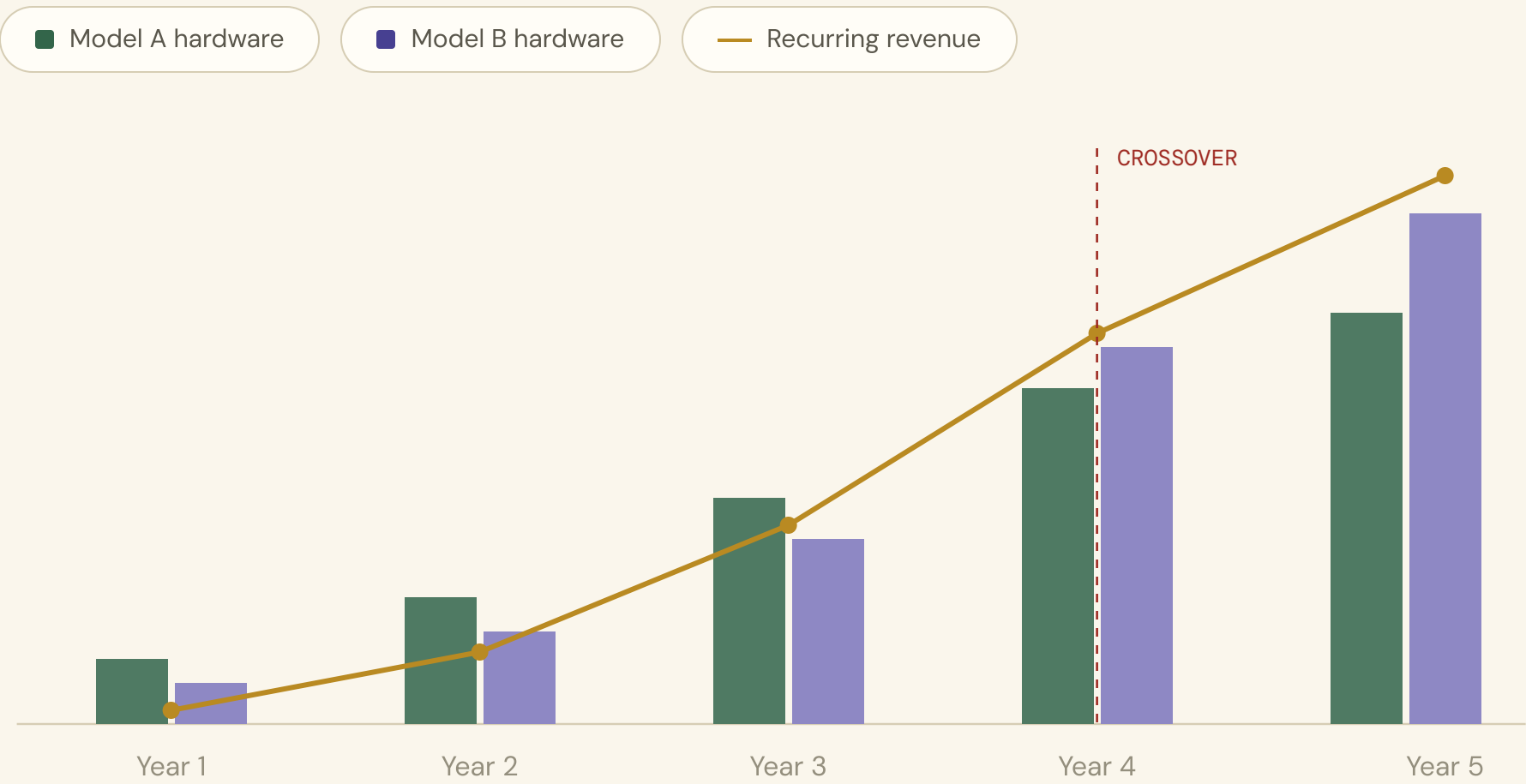
COST PER MW, VIA OEM CLOUD

[]

Cloud onboarding carries no hardware cost, so blended cost per MW falls as that path scales.

Recurring revenue crosses hardware.

Shape is illustrative. Every value below is an unfilled placeholder pending the model.



The crossover is the story of this slide. Where it lands is what the model has to prove.

DRIVER	Y1	Y2	Y3	Y4	Y5
MODEL A UNITS	[]	[]	[]	[]	[]
MODEL B UNITS	[]	[]	[]	[]	[]
ASSETS UNDER MGMT	[]	[]	[]	[]	[]
MW UNDER MGMT	[]	[]	[]	[]	[]
HARDWARE REVENUE	[]	[]	[]	[]	[]
RECURRING REVENUE	[]	[]	[]	[]	[]
GROSS MARGIN %	[]	[]	[]	[]	[]
BURN	[]	[]	[]	[]	[]

A live channel into DISCOMs.

Two industrial partners and a channel onto three continents, run at roughly \$2,000 a month.



HARDWARE

Custom PCB in fabrication with PCB Power India



LIVE TELEMETRY

Kintech Synergy BESS streaming today



CHANNEL

5 DISCOMs via Beckn DEG, across 3 continents



DEPLOYED

2 operating plants live since December 2025

Zodiac Energy

2,500+ sites, 30+ MW rooftop solar in Gujarat. Data sharing live, generation records feeding model training, gateway telemetry pilot in setup.

NSE & BSE LISTED

Kintech Synergy

Thirty years in solar, wind and storage. Live BESS assets streaming gateway telemetry, with state-of-charge and dispatch validation running.

₹800 CR+ REVENUE EPC

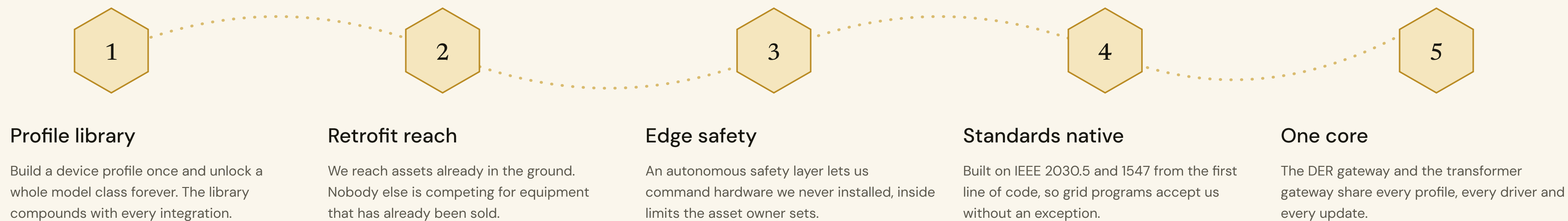
DEVICE PROFILES BUILT
[X]

SITES LIVE
[X]

MW UNDER MANAGEMENT
[X]

IEEE 2030.5
[status]

Five advantages that compound.



Each profile built and each site retrofit widens a lead nobody else is chasing.

Built by practitioners.

FOUNDER



Arham Shah

FOUNDER & CEO

M.S. Energy, Stanford · B.S. Industrial Engineering, UIUC

Ex-Tesla, Rivian, Beckn. Launched the first open energy networks in India and Brazil.

ADVISORS



Sujith Nair

Co-founder, Beckn & FIDE



Dr. Pramod Varma

Chief architect of UPI and Aadhaar

ENGINEERING



Sudheer Kumar

Founding forward-deployment engineer



Yuvaraju Meenuga

Founding backend and integrations engineer

15+ years combined at Uplight and AutoGrid

Three full-time, zero attrition

Production PCB already in fabrication

\$65,000 raised to date · ~\$2,000 monthly burn

The team that shipped the pilot is the team building the company.

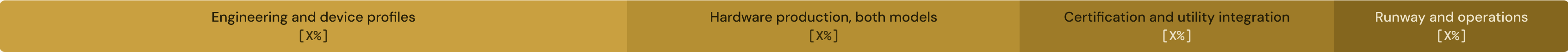
THE ASK

\$1.5M.

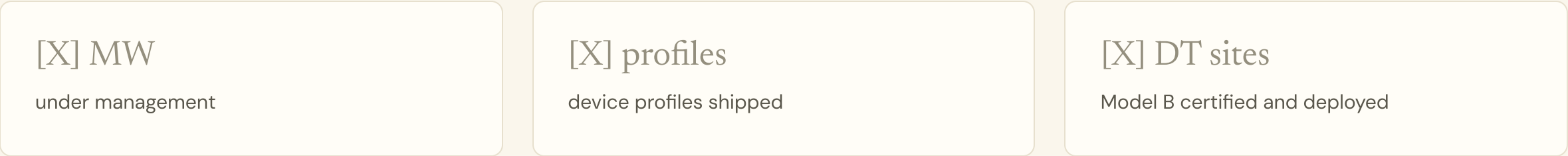


On a SAFE, [\$ cap] cap.

USE OF FUNDS



THIS GETS US TO



Runway [X] months.

Everyone else is chasing next year's assets. We're switching on the ones already in the ground.